Exp No:3.a Conduct an experiment to show data visualization using line plot

AIM:
To perform data visualization using the line plot.

Description: Take any sample data either through csv file or data fetched directly through code.

CODE:

```
 import matplotlib.pyplot as sales

Months = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct"]

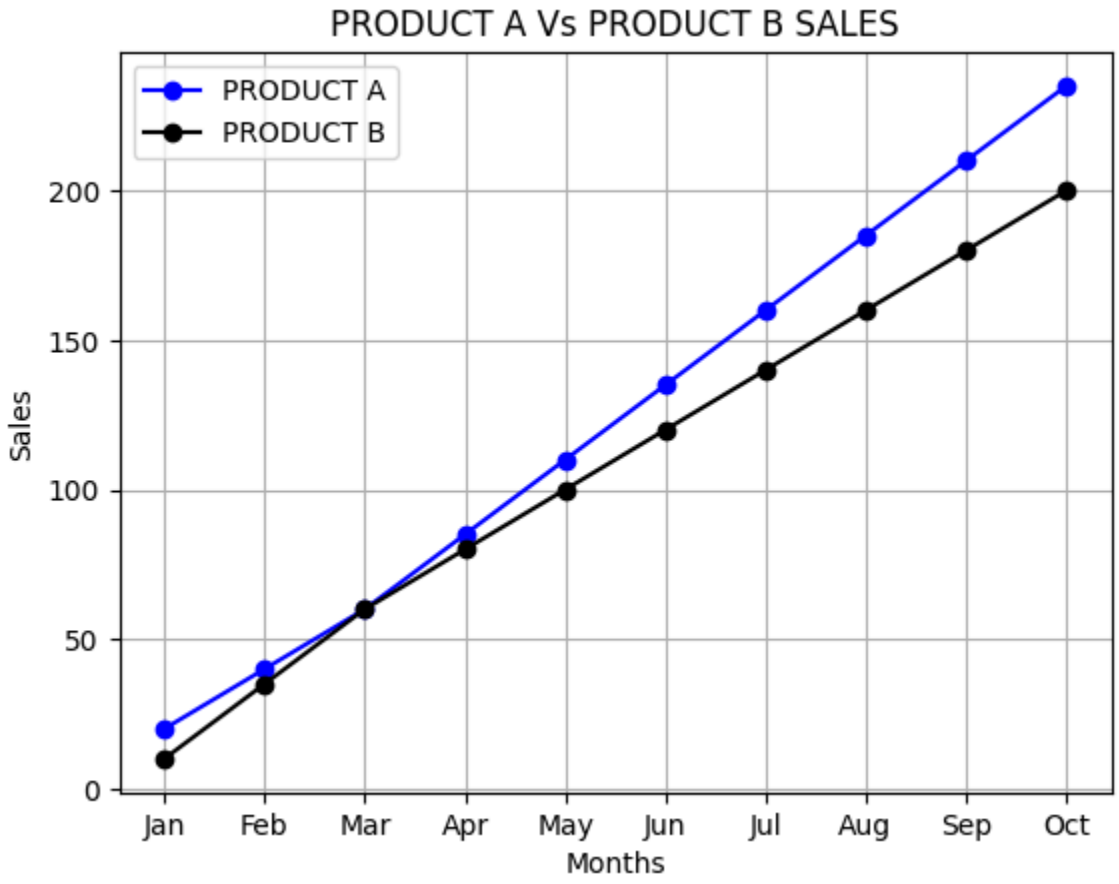
Product_A = [20, 40, 60, 85, 110, 135, 160, 185, 210, 235]
Product_B = [10, 35, 60, 80, 100, 120, 140, 160, 180, 200]

sales.plot(Months, Product_A, color="blue", marker="o", label="PRODUCT A")
sales.plot(Months, Product_B, color="black", marker="o", label="PRODUCT
B")

sales.title("PRODUCT A Vs PRODUCT B SALES")
sales.xlabel("Months")
sales.ylabel("Sales")
sales.legend(loc="upper left")
sales.grid(True)

sales.show()
```

SAMPLE OUTPUT:



RESULT:

The data visualization using the line plot is done successfully

Exp No:3.b Conduct an experiment to show data visualization using bar chart

AIM:
To perform data visualization using the bar chart.

Description: Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as marks
import numpy as np

Subjects = ['Maths', 'Physics', 'Chemistry', 'English', 'Computer',
'Biology']

xaxis = np.arange(len(Subjects))

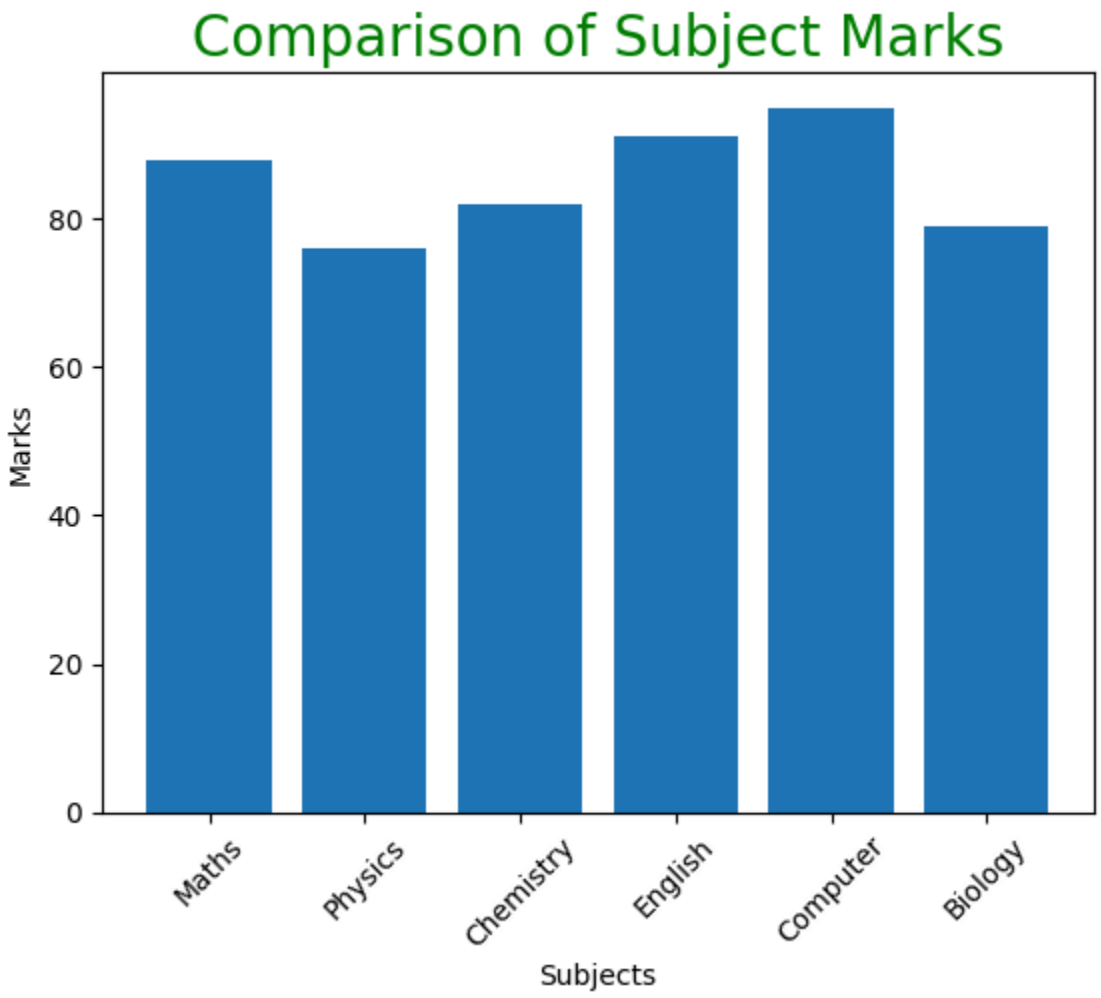
Marks = [88, 76, 82, 91, 95, 79]

marks.bar(Subjects, Marks)

marks.xticks(xaxis, Subjects, rotation=45)
marks.xlabel("Subjects")
marks.ylabel("Marks")
marks.title("Comparison of Subject Marks", fontsize=20, color="green")

marks.show()
```

SAMPLE OUTPUT:



RESULT:

The Data Visualization for the given dataset using the bar chart is done successfully.

Exp No:3.c Conduct an experiment to show data visualization using pie chart

AIM:
To analyse and visualize the given dataset using the pie chart.

Description: Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as expense

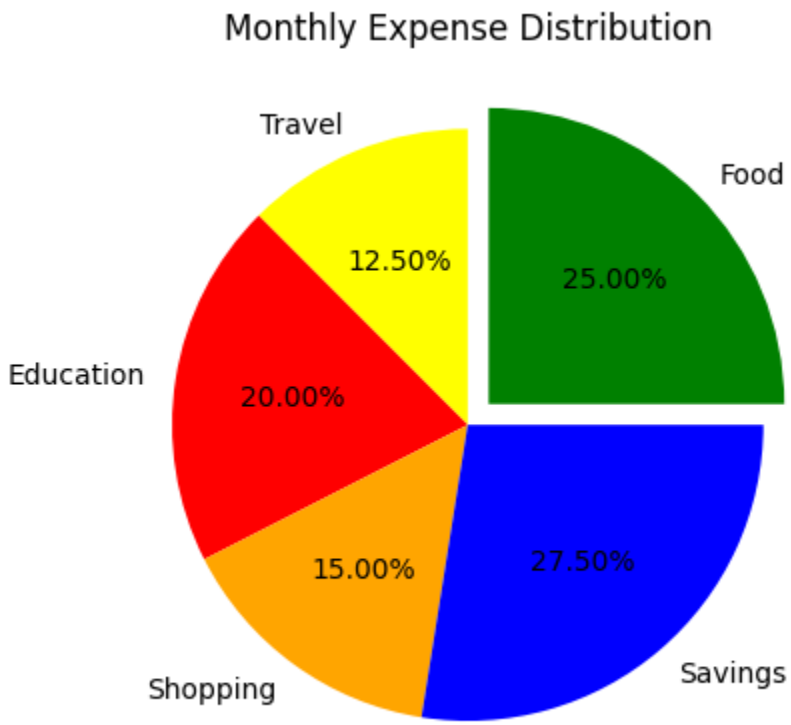

labels = ['Food', 'Travel', 'Education', 'Shopping', 'Savings']
Amount = [5000, 2500, 4000, 3000, 5500]

colors = ['green', 'yellow', 'red', 'orange', 'blue']
explode = (0.1, 0, 0, 0, 0)

expense.pie(Amount, labels=labels, colors=colors,
            explode=explode, autopct='%0.2f%%')

expense.title('Monthly Expense Distribution')
expense.show()
```

SAMPLE OUTPUT:



RESULT:

The data visualization of the given dataset is done using the pie chart.